SQUAMOUS CELL CARCINOMA

AT A GLANCE

- One of the most common human malignancies, along with basal cell carcinoma.
- Diagnosis confirmed by biopsy.
- Primarily caused by ultraviolet (UV) radiation.
- Precursor lesions include actinic keratosis and Bowen disease.
- Treatment options include surgical excision, Mohs microscopically controlled surgery, and radiation therapy.

ETIOLOGY AND PATHOGENESIS

Predisposing Factors

Multiple acquired and genetic factors can predispose individuals to squamous cell carcinoma (SCC). Patients often have a combination of risk factors that collectively contribute to SCC development. For example, a single skin site may be exposed to both UV radiation and another environmental carcinogen.

Precursor Lesions

Most SCCs arise from precursor lesions such as **actinic keratoses (AKs)** or **Bowen disease** (squamous cell carcinoma in situ).

Ultraviolet Radiation Exposure

UV radiation is the predominant risk factor for SCC. There is a well-established linear correlation between cumulative UV exposure and SCC incidence. The incidence of SCC approximately doubles with each 8- to 10-degree decrease in geographic latitude, being highest near the equator.

Epidemiological evidence supports this: - World War II veterans stationed in the Pacific had significantly higher rates of SCC compared to those stationed in Europe. - Japanese individuals who emigrated to Hawaii show higher SCC rates than those who remained in Japan.

UV radiation appears more strongly linked to SCC development than to basal cell carcinoma (BCC): - SCC rates increase more rapidly than BCC rates with increasing UV exposure. - In mouse models, UV-induced skin cancers are almost exclusively SCCs. - Patients undergoing long-term psoralen plus ultraviolet A (PUVA) therapy for psoriasis have a 30-fold increased risk of non-melanoma skin cancer, predominantly SCC.

Ionizing Radiation

Exposure to ionizing radiation is strongly associated with SCC development. This association is particularly evident in individuals with fair skin who are prone to sunburn.

Environmental Carcinogens

Occupational and environmental exposure to carcinogens such as **arsenic**, **aromatic hydrocarbons**, **insecticides**, and **herbicides** increases SCC risk. Most chemical carcinogens—except for 3-methylcholanthrene and anthramine—preferentially induce SCC rather than BCC. Additionally, **tobacco use** and **alcohol consumption** are major risk factors for SCC of the oral cavity.

Immunosuppression

Chronic immunosuppression significantly increases the risk of SCC, especially on sun-exposed skin. - Renal transplant recipients have an 18-fold higher risk of developing SCC. - These cancers typically appear 3 to 7 years after initiation of immunosuppressive therapy, with drugs such as **corticosteroids**, **azathioprine**, and **cyclosporine** implicated. - With the growing number of organ transplant recipients, SCC management in immunosuppressed patients has become increasingly important. - Patients with hematologic malignancies such as leukemia and lymphoma also exhibit increased incidence and more aggressive forms of SCC.

Other Predisposing Conditions

- Scars (especially chronic or burn-related)
- Long-term heat exposure (e.g., thermal keratoses from repeated heat application)
- Chronic inflammatory or scarring dermatoses (e.g., discoid lupus, osteomyelitis sinuses)
- **Human papillomavirus (HPV)** infection, particularly in immunocompromised individuals
- **Genodermatoses**:
- Albinism
- Xeroderma pigmentosum
- Porokeratosis
- Epidermolysis bullosa

Genetic Factors

Skin pigmentation plays a role in SCC susceptibility. Individuals with intermediate pigmentation, such as Asians and Polynesians, have intermediate SCC risk. The **MC1R gene**, which encodes the melanocortin 1 receptor and plays a key role in melanogenesis, is a major determinant of skin and hair pigmentation. It is highly polymorphic, with over 20 known variants. Certain **variant MC1R alleles** are associated with an increased risk of SCC independent of skin type or hair color, suggesting a direct role in carcinogenesis beyond pigmentation.